

Independent Wolfram Language Verification of KL-Derived Linear Systems from `LPv3.tex`

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1 Introduction

The companion note `LPv3.tex` derives parametric solutions of the Knill–Laflamme (KL) Z -marginal conditions for six quantum-code examples (five one-parameter families A–E and a worked $((5, 2, 2))$ example) using Python/SymPy. This document reports an independent, end-to-end re-derivation and verification of *every* numerical and algebraic claim in `LPv3.tex`, carried out entirely in Wolfram Language (Mathematica). The accompanying script `verify_LP.wl` (425 lines) is self-contained and can be run via

`wolframscript -file verify_LP.wl`

Result. *Zero discrepancies.* All parametric solutions, feasibility regions, Z -marginal vectors, Σ_Z polynomials, Σ_Z ranges, and convex-hull dimension tests match exactly.

1.1 Scope of verification

The verification pipeline, applied to each of the six examples, checks:

1. Build the sign matrix $s_i(x) = (-1)^{x_i}$ for each codeword.
2. Construct the augmented matrix $[A \mid b]$ encoding Z -marginal equalities and normalization.
3. Compute **RowReduce** (RREF over $\mathbb{Q}$) to get the affine parametric solution.
4. Identify pivot/free columns, extract the parametric solution.
5. Impose nonnegativity via **Reduce** to obtain the feasibility region.
6. Compute Z -marginal vectors from *both* C_0 and C_1 sides and confirm they match.
7. Compute $\Sigma_Z = \sum_i \langle Z_i \rangle^2$ and its range via **Minimize/Maximize**.
8. (Appendix A.3 test) Minimize and maximize each $\langle Z_i \rangle$ to determine whether $\text{conv}(V_0) \cap \text{conv}(V_1)$ is a point or has positive dimension.

1.2 Relationship to the Lean formalization

Families A–E and the $(5, 2, 2)$ example are also formalized in Lean 4 (Mathlib). The Lean proofs verify that the *given* probability assignments satisfy the SS condition, residue-shift screen, normalization, and Z -type KL equalities—that is, **correctness**: “these formulas work.” The Wolfram verification additionally confirms **completeness**: “these are *all* solutions,” by deriving the parametric families from scratch via RREF and showing the feasibility regions are exactly as claimed.

Property	Lean	Wolfram
SS support condition	✓	—
Residue-shift screen	✓	—
Probability normalization	✓	✓
Z -type KL ($\langle Z_i \rangle$ matches target)	✓	✓
Σ_Z range	✓	✓
Parametric solution is complete (RREF)	—	✓
Feasibility region is exact	—	✓
Z -marginals from C_0 and C_1 agree	—	✓
$\text{conv}(V_0) \cap \text{conv}(V_1)$ dimension	—	✓

2 Summary of results

Table 1 gives the headline numbers. Full per-family comparisons follow in Section 3.

3 Family-by-family verification

For each family we list the supports, the parametric solution obtained from **RowReduce**, the feasibility region from **Reduce**, the Z -marginal vector, Σ_Z and its range, and the comparison with **LPv3.tex**. In every case the results agree exactly.

3.1 Family A: $(m, \mathbf{w}, S_0, S_1) = (6, (4, 5, 1, 2, 2), 0, 3)$

Supports. $C_0 = \{10010, 11101, 01100\}$, $C_1 = \{11000, 10111, 00110\}$.

Example	Vars	Rank	Free	Feasibility	Σ_Z range	conv test
Family A	6	5	1	$0 \leq t \leq \frac{1}{2}$	$[\frac{1}{2}, 1]$	segment
Family B	8	7	1	$0 \leq t \leq \frac{1}{3}$	$[\frac{1}{3}, \frac{5}{9}]$	segment
Family C	8	7	1	$0 \leq t \leq \frac{1}{3}$	$[\frac{1}{3}, \frac{5}{9}]$	segment
Family D	8	7	1	$0 \leq t \leq \frac{1}{3}$	$[\frac{1}{3}, \frac{5}{9}]$	segment
Family E	6	5	1	$0 \leq t \leq \frac{1}{2}$	$[\frac{1}{2}, 1]$	segment
Worked (5, 2, 2)	9	7	2	triangle	$\{\frac{3}{7}\}$	point

Table 1: Summary of verification results. “Vars” is $|C_0| + |C_1|$; “Rank” is the rank of the augmented system after RREF; “Free” is the number of free parameters. For the worked example, the feasibility region is $u \geq 0, v \geq 0, u + v \leq \frac{2}{7}$. All entries match `LPv3.tex`.

Linear system. 6 variables, rank 5, 1 free parameter. Free variable: $t := Y_{00110}$.

Parametric solution (Wolfram output).

$$\begin{aligned} X_{10010} &= \frac{1}{2}, & X_{11101} &= \frac{1}{2} - t, & X_{01100} &= t, \\ Y_{11000} &= \frac{1}{2}, & Y_{10111} &= \frac{1}{2} - t, & Y_{00110} &= t. \end{aligned}$$

Feasibility. $0 \leq t \leq \frac{1}{2}$. ✓

Z-marginals. Computed independently from both C_0 and C_1 :

$$(\langle Z_1 \rangle, \dots, \langle Z_5 \rangle) = (2t - 1, 0, 0, 0, 2t). \quad \checkmark$$

Σ_Z . $\Sigma_Z(t) = 8t^2 - 4t + 1$, range $[\frac{1}{2}, 1]$, minimum at $t = \frac{1}{4}$. ✓

3.2 Family B: $(m, \mathbf{w}, S_0, S_1) = (6, (1, 1, 1, 5, 3), 0, 2)$

Supports. $C_0 = \{00000, 00110, 10010, 11101\}$, $C_1 = \{00011, 01100, 10100, 11000\}$.

Linear system. 8 variables, rank 7, 1 free parameter. Free variable: $t := Y_{11000}$.

Parametric solution.

$$\begin{aligned} X_{00000} &= X_{11101} = Y_{00011} = Y_{10100} = \frac{1}{3}, \\ X_{00110} &= Y_{01100} = \frac{1}{3} - t, \\ X_{10010} &= Y_{11000} = t. \end{aligned}$$

Feasibility. $0 \leq t \leq \frac{1}{3}$. ✓

Z-marginals.

$$(\langle Z_1 \rangle, \dots, \langle Z_5 \rangle) = (\frac{1}{3} - 2t, \frac{1}{3}, -\frac{1}{3} + 2t, \frac{1}{3}, \frac{1}{3}). \quad \checkmark$$

Σ_Z . $\Sigma_Z(t) = 8t^2 - \frac{8}{3}t + \frac{5}{9}$, range $[\frac{1}{3}, \frac{5}{9}]$, minimum at $t = \frac{1}{6}$. ✓

3.3 Family C: $(m, \mathbf{w}, S_0, S_1) = (6, (3, 3, 1, 1, 1), 0, 4)$

Supports. $C_0 = \{00000, 01111, 10111, 11000\}$, $C_1 = \{01001, 01100, 10010, 10100\}$.

Linear system. 8 variables, rank 7, 1 free parameter. Free variable: $t := Y_{10100}$.

Parametric solution.

$$\begin{aligned} X_{00000} &= X_{11000} = Y_{01001} = Y_{10010} = \frac{1}{3}, \\ X_{01111} &= Y_{01100} = \frac{1}{3} - t, \\ X_{10111} &= Y_{10100} = t. \end{aligned}$$

Feasibility. $0 \leq t \leq \frac{1}{3}$. ✓

Z-marginals.

$$(\langle Z_1 \rangle, \dots, \langle Z_5 \rangle) = (\frac{1}{3} - 2t, -\frac{1}{3} + 2t, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}). \quad \checkmark$$

Σ_Z . $\Sigma_Z(t) = 8t^2 - \frac{8}{3}t + \frac{5}{9}$, range $[\frac{1}{3}, \frac{5}{9}]$, minimum at $t = \frac{1}{6}$. ✓

3.4 Family D: $(m, \mathbf{w}, S_0, S_1) = (6, (3, 1, 3, 5, 5), 0, 4)$

Supports. $C_0 = \{00000, 01001, 11101, 11110\}$, $C_1 = \{00011, 01100, 10111, 11000\}$.

Linear system. 8 variables, rank 7, 1 free parameter. Free variable: $t := Y_{10111}$.

Parametric solution.

$$\begin{aligned} X_{00000} &= X_{11110} = Y_{01100} = Y_{11000} = \frac{1}{3}, \\ X_{01001} &= Y_{00011} = \frac{1}{3} - t, \\ X_{11101} &= Y_{10111} = t. \end{aligned}$$

Feasibility. $0 \leq t \leq \frac{1}{3}$. ✓

Z-marginals.

$$(\langle Z_1 \rangle, \dots, \langle Z_5 \rangle) = (\frac{1}{3} - 2t, -\frac{1}{3}, \frac{1}{3} - 2t, \frac{1}{3}, \frac{1}{3}). \quad \checkmark$$

Σ_Z . $\Sigma_Z(t) = 8t^2 - \frac{8}{3}t + \frac{5}{9}$, range $[\frac{1}{3}, \frac{5}{9}]$, minimum at $t = \frac{1}{6}$. ✓

3.5 Family E: $(m, \mathbf{w}, S_0, S_1) = (8, (1, 6, 2, 5, 5), 0, 4)$

Supports. $C_0 = \{01011, 10110, 10101\}$, $C_1 = \{00111, 11010, 11001\}$.

Linear system. 6 variables, rank 5, 1 free parameter. Free variable: $t := Y_{11001}$.

Parametric solution.

$$\begin{aligned} X_{01011} &= Y_{00111} = \frac{1}{2}, \\ X_{10110} &= Y_{11010} = \frac{1}{2} - t, \\ X_{10101} &= Y_{11001} = t. \end{aligned}$$

Feasibility. $0 \leq t \leq \frac{1}{2}$. ✓

Z-marginals.

$$(\langle Z_1 \rangle, \dots, \langle Z_5 \rangle) = (0, 0, 0, -1 + 2t, -2t). \quad \checkmark$$

Σ_Z . $\Sigma_Z(t) = 8t^2 - 4t + 1$, range $[\frac{1}{2}, 1]$, minimum at $t = \frac{1}{4}$. ✓

3.6 Worked example: $((5, 2, 2)), m = 7$

Supports. $C_0 = \{00000, 01111, 10111\}$, $C_1 = \{00011, 00101, 00110, 11001, 11010, 11100\}$.

Linear system. 9 variables, rank 7, 2 free parameters. Free variables: $u := Y_{11010}$, $v := Y_{11100}$.

Parametric solution.

$$\begin{aligned} X_{00000} &= \frac{3}{7}, & X_{01111} &= X_{10111} = \frac{2}{7}, \\ Y_{00011} &= \frac{1}{7} + v, & Y_{00101} &= \frac{1}{7} + u, & Y_{00110} &= \frac{3}{7} - u - v, \\ Y_{11001} &= \frac{2}{7} - u - v, & Y_{11010} &= u, & Y_{11100} &= v. \end{aligned}$$

Feasibility. $u \geq 0$, $v \geq 0$, $u + v \leq \frac{2}{7}$. ✓

Z-marginals. Constant (independent of u, v):

$$(\langle Z_1 \rangle, \dots, \langle Z_5 \rangle) = \left(\frac{3}{7}, \frac{3}{7}, -\frac{1}{7}, -\frac{1}{7}, -\frac{1}{7} \right). \quad \checkmark$$

$$\Sigma_Z. \quad \Sigma_Z = 2\left(\frac{3}{7}\right)^2 + 3\left(\frac{1}{7}\right)^2 = \frac{3}{7} \text{ (constant). } \checkmark$$

4 Convex-hull dimension test (Appendix A.3 of LPv3.tex)

The “practical test for varying marginals” from Appendix A.3 of LPv3.tex is: for each example, minimize and maximize each Z -marginal coordinate $\langle Z_i \rangle$ subject to all feasibility constraints (equalities from RREF plus nonnegativity). If $\min \langle Z_i \rangle \neq \max \langle Z_i \rangle$ for any i , then $\text{conv}(V_0) \cap \text{conv}(V_1)$ has positive dimension (marginals vary); if all are equal, the intersection is a single point.

We implemented this using **Minimize/Maximize** on the parametric Z -marginal expressions subject to the **Reduce**-derived feasibility region.

Example	$\langle Z_1 \rangle$	$\langle Z_2 \rangle$	$\langle Z_3 \rangle$	$\langle Z_4 \rangle$	$\langle Z_5 \rangle$	Verdict
Family A	$[-1, 0]$	$\{0\}$	$\{0\}$	$\{0\}$	$[0, 1]$	varies
Family B	$[-\frac{1}{3}, \frac{1}{3}]$	$\{\frac{1}{3}\}$	$[-\frac{1}{3}, \frac{1}{3}]$	$\{\frac{1}{3}\}$	$\{\frac{1}{3}\}$	varies
Family C	$[-\frac{1}{3}, \frac{1}{3}]$	$[-\frac{1}{3}, \frac{1}{3}]$	$\{\frac{1}{3}\}$	$\{\frac{1}{3}\}$	$\{\frac{1}{3}\}$	varies
Family D	$[-\frac{1}{3}, \frac{1}{3}]$	$\{-\frac{1}{3}\}$	$[-\frac{1}{3}, \frac{1}{3}]$	$\{\frac{1}{3}\}$	$\{\frac{1}{3}\}$	varies
Family E	$\{0\}$	$\{0\}$	$\{0\}$	$[-1, 0]$	$[-1, 0]$	varies
Worked	$\{\frac{3}{7}\}$	$\{\frac{3}{7}\}$	$\{-\frac{1}{7}\}$	$\{-\frac{1}{7}\}$	$\{-\frac{1}{7}\}$	fixed

Table 2: Dimension test: ranges of individual Z -marginal coordinates over the feasibility region. Intervals denote varying coordinates; singletons denote fixed ones. All agree with LPv3.tex.

Interpretation.

- **Families A–E:** $\text{conv}(V_0) \cap \text{conv}(V_1)$ is a line segment in $\mathbb{R}^5$, parameterized by t . Two of the five $\langle Z_i \rangle$ coordinates vary (the rest are constant); the varying pair always involves the coordinates whose sign vectors differ across the free-parameter codeword pair.
- **Worked** $((5, 2, 2))$: $\text{conv}(V_0) \cap \text{conv}(V_1)$ is a single point $(\frac{3}{7}, \frac{3}{7}, -\frac{1}{7}, -\frac{1}{7}, -\frac{1}{7})$. The two free parameters (u, v) only change how this fixed point is decomposed as a convex combination of vertices in V_1 ; they do not move the marginal vector itself.

5 Structural observations

The Wolfram computation confirms the following structural patterns noted in LPv3.tex and lambdav2.tex.

1. **Families A and E** (3+3 supports) share the same Σ_Z polynomial $8t^2 - 4t + 1$, with range $[\frac{1}{2}, 1]$ over the respective feasibility interval.

2. **Families B, C, and D** (4+4 supports) share the same Σ_Z polynomial $8t^2 - \frac{8}{3}t + \frac{5}{9}$, with range $[\frac{1}{3}, \frac{5}{9}]$.
3. For all five one-parameter families, the Z -marginal vector varies continuously with t , confirming that $\text{conv}(V_0) \cap \text{conv}(V_1)$ is a line segment (not a point).
4. For the worked $((5, 2, 2))$ example, the two free parameters produce a 2-simplex (triangle) in probability space, but the Z -marginals are constant, so $\text{conv}(V_0) \cap \text{conv}(V_1)$ collapses to a single point.
5. All computations are exact over $\mathbb{Q}$ —no floating point is used at any stage.

6 Method details

6.1 Core operation: RowReduce

The central computation is **RowReduce** on the augmented matrix $[A \mid b]$ over $\mathbb{Q}$. The matrix has $n + 2$ rows (n shared-marginal equations plus 2 normalizations) and $|C_0| + |C_1| + 1$ columns (one per probability variable, plus the RHS). The (i, j) entry in the marginal block is $(-1)^{(x_j)_i}$ for C_0 -columns and $-(-1)^{(y_k)_i}$ for C_1 -columns.

After RREF, pivot columns identify determined variables and free columns identify parameters. The parametric solution is read off directly from the reduced rows.

6.2 Feasibility via Reduce

Substituting the parametric expressions into $p_j \geq 0$ for every probability variable yields a conjunction of linear inequalities in the free parameters. Calling **Reduce**[..., **freeParams**, **Reals**] produces a simplified description. For the one-parameter families, this is always an interval $[0, t_{\max}]$. For the two-parameter worked example, **Reduce** returns a case-split form; equivalence with the simpler triangle description $u \geq 0$, $v \geq 0$, $u + v \leq \frac{2}{7}$ was confirmed using **Resolve**[**Exists**[...], **Reals**].

6.3 Z -marginals and cross-check

The Z -marginal vector is computed independently from both the C_0 and C_1 sides:

$$\langle Z_i \rangle_{C_0} = \sum_{x \in C_0} X_x (-1)^{x_i}, \quad \langle Z_i \rangle_{C_1} = \sum_{y \in C_1} Y_y (-1)^{y_i}.$$

In every example, **Simplify**[**zVec0** - **zVec1**] returns the zero vector, confirming the shared-marginal equations are satisfied.

6.4 Optimization

Σ_Z ranges are obtained by calling **Minimize** and **Maximize** on $\sum_i \langle Z_i \rangle^2$ subject to the feasibility region. The same approach is used for the dimension test (Section 4), applied to individual $\langle Z_i \rangle$ coordinates.

A Wolfram Language code listing

The complete verification script is in `verify_LP.wl` (425 lines). Below we reproduce the core reusable functions; the full file additionally contains the per-example invocations and the dimension test from Section 4.

A.1 Sign vector construction

```
bitToSign[s_String] := Table[
  If[StringPart[s, i] == "0", 1, -1],
  {i, StringLength[s]}
]
```

A.2 Build augmented matrix and solve via RREF

```
solveKL[c0Strings_List, c1Strings_List] := Module[
  {n, n0, n1, nVars, sC0, sC1, aug, rref,
   pivotCols, freeCols, varNames, pos, p},

  n = StringLength[c0Strings[[1]]];
  n0 = Length[c0Strings];
  n1 = Length[c1Strings];
  nVars = n0 + n1;

  sC0 = bitToSign /@ c0Strings;
  sC1 = bitToSign /@ c1Strings;

  (* Augmented matrix [A | b]:
     Rows 1..n: Z_i shared-marginal equations
     Row n+1: sum X_j = 1
     Row n+2: sum Y_k = 1 *)
  aug = Table[
    Join[Table[sC0[[j, i]], {j, n0}],
         Table[-sC1[[k, i]], {k, n1}], {0}],
    {i, n}];
  AppendTo[aug,
    Join[ConstantArray[1, n0], ConstantArray[0, n1], {1}]];
  AppendTo[aug,
    Join[ConstantArray[0, n0], ConstantArray[1, n1], {1}]];

  rref = RowReduce[aug];

  (* Identify pivots *)
  pivotCols = {};
  Do[pos = Position[rref[[row]], _?(# != 0 &)];
    If[Length[pos] > 0,
      p = pos[[1, 1]];
      If[p <= nVars, AppendTo[pivotCols, p]],
      {row, Length[rref]}];
  freeCols = Complement[Range[nVars], pivotCols];

  varNames = Join[("X_" <> # &) /@ c0Strings,
                  ("Y_" <> # &) /@ c1Strings];

  <| "n" -> n, "n0" -> n0, "n1" -> n1,
    "nVars" -> nVars, "rref" -> rref,
    "pivotCols" -> pivotCols, "freeCols" -> freeCols,
    "varNames" -> varNames,
    "rank" -> Length[pivotCols],
    "nFree" -> Length[freeCols],
    "C0" -> c0Strings, "C1" -> c1Strings,
    "sC0" -> sC0, "sC1" -> sC1 |>
]
```

A.3 Extract parametric solution

```
extractSolution[result_Association] := Module[
  {rref, nVars, pivotCols, freeCols, varNames,
   freeSyms, sol, row, pc, expr},
```

```

rref = result[['rref']; nVars = result[['nVars'];
pivotCols = result[['pivotCols'];
freeCols = result[['freeCols'];
varNames = result[['varNames'];

freeSyms = Table[Symbol["t" <> ToString[k]],
                 {k, Length[freeCols]}}];
sol = <||>;

(* Free variables *)
Do[sol[varNames[[freeCols[[k]]]]] = freeSyms[[k]],
   {k, Length[freeCols]}}];

(* Pivot variables from RREF *)
Do[row = rref[[r]]; pc = pivotCols[[r]];
  If[pc <= nVars,
    expr = row[[nVars + 1]];
    Do[expr = expr - row[[freeCols[[k]]]] * freeSyms[[k]],
       {k, Length[freeCols]}}];
    sol[varNames[[pc]]] = Simplify[expr],
    {r, Length[pivotCols]}}];

<| 'solution' -> sol, 'freeSyms' -> freeSyms,
   'freeNames' -> (varNames[[#]] & /@ freeCols) |>
]

```

A.4 Compute Z-marginals and Σ_Z

```

computeZMarginals[result_Association,
                  solData_Association] := Module[
{n, sC0, sC1, c0, c1, sol,
 probs0, probs1, zVec0, zVec1, sigZ},

n = result["n"]; sC0 = result["sC0"];
sC1 = result["sC1"];
c0 = result["C0"]; c1 = result["C1"];
sol = solData["solution"];

probs0 = Table[sol["X_" <> c0[[j]]], {j, Length[c0]};
probs1 = Table[sol["Y_" <> c1[[k]]], {k, Length[c1]};

zVec0 = Table[
  Sum[sC0[[j, i]] * probs0[[j]], {j, Length[c0]},
  {i, n}] // Simplify;
zVec1 = Table[
  Sum[sC1[[k, i]] * probs1[[k]], {k, Length[c1]},
  {i, n}] // Simplify;

sigZ = Total[zVec0^2] // Expand // Simplify;

<| 'zMargC0' -> zVec0, 'zMargC1' -> zVec1,
   'zMargMatch' ->
   (Simplify[zVec0 - zVec1] ==
    ConstantArray[0, n]),
   'sigmaZ' -> sigZ |>
]

```


A.5 Feasibility region

```
computeFeasibility[result_Association,
  solData_Association] := Module[
  {varNames, sol, freeSyms, allExprs, ineqs},

  varNames = result['varNames'];
  sol = solData['solution'];
  freeSyms = solData['freeSyms'];

  allExprs = Table[sol[vn], {vn, varNames}];
  ineqs = And @@ ((# >= 0) & /@ allExprs);

  If[Length[freeSyms] == 1,
    Reduce[ineqs, freeSyms[[1]], Reals],
    Reduce[ineqs, freeSyms, Reals]]
]
```

A.6 Example invocation (Family A)

```
result = solveKL[{"10010'", "11101'", "01100"},
  {"11000'", "10111'", "00110"}];
solData = extractSolution[result];
zData = computeZMarginals[result, solData];
region = computeFeasibility[result, solData];

(* SigmaZ range *)
t1 = solData['freeSyms'][[1]];
Minimize[{zData['sigmaZ'], region}, t1]
Maximize[{zData['sigmaZ'], region}, t1]
```

Conclusion

Every algebraic and numerical claim in `LPv3.tex`—parametric solutions, feasibility regions, Z -marginal vectors, Σ_Z polynomials and their ranges, and the convex-hull dimension classification—has been independently verified using exact rational arithmetic in the Wolfram Language. The verification covers all six examples (Families A–E and the worked $((5, 2, 2))$ example). Zero discrepancies were found.